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Integrating Fragmented Risk Knowledge: Sheaf Theory for Risk Analysts

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ABSTRACT

As sociotechnical systems grow more interconnected, data-rich, and intricately distributed across organizational and disciplinary boundaries and over levels of organizational and technology hierarchies, there is an increasing need for risk assessment and risk management modeling tools that describe how local information, constraints, and decisions interact to shape global outcomes. "Systems-of-systems" (SoS) provide a starting point, but additional methods are needed to understand how risk assessments and management decisions based on partial, local knowledge can (or cannot) be integrated into globally coherent strategies. This paper explores applications of a powerful branch of mathematics, *sheaf theory*, as a mathematically principled framework for reasoning about local-to-global relationships, structural consistency, and context-dependent interpretation and integration of risk-related information and messages in complex systems. We seek to present an accessible, risk analyst-oriented introduction to core technical concepts and to illustrate the practical value of these mathematical techniques for risk analysis through examples drawn from risk psychology and decision-making under uncertainty; fusion of sensor data and other partial information; causal modeling and quantitative risk assessment of complex engineering and biological systems; risk communication and Social Amplification of Risk (SARF); collective and distributed planning and policy coordination; and ongoing monitoring and evaluation of threats and intervention outcomes. We show how sheaf-theoretic methods can detect irreconcilable policy conflicts, identify gaps in information fusion, simulate belief propagation under constraints, and support modular, multi-level modeling. Applications span risk communication, environmental monitoring, emergency planning, and regulatory governance. We conclude with a discussion of how sheaf theory might be integrated into mainstream risk science.

1 | Introduction: The Risk of Fragmented Information

Many of the most damaging failures in risk management, such as conflicting public health advisories, uncoordinated emergency evacuations, regulatory contradictions, and the rapid spread of disinformation, share a common root cause: Fragmented knowledge. Different agents, institutions, or systems operate with partial, incompatible, or misleading information. Even when each part behaves reasonably based on its own local data or beliefs, the system as a whole may fail to reach a coherent or desirable outcome.

This paper introduces a powerful and general mathematical framework, *sheaf theory*, for analyzing such failures. It is a long paper because we believe that the interlinked concepts in sheaf theory cast useful light on many aspects of risk analysis in a wide variety of settings, from individual and team decision

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Apropos of Shafer: "Probability really isn't about the numbers; it's about the structure of reasoning". (Cited in Pearl, 1988, https://doi.org/10.1016/C2009-0-27609-4)

Wicked problems?

optimization to organizational, institutional, and societal risk assessment and management decision-making. But the most essential idea can be summarized succinctly: Risk management is made challenging in most real-world applications by the fact that people have fragmented views of risk, rather than everyone agreeing on risk models and on risk management goals and worthwhile trade-offs. Sheaf theory can help to describe such fragmented views, ascertain whether they can be integrated, and, if not, identify needed changes in information flows to restore the possibility of an integrated view of risk and a coordinated response to it. We call this the *sheaf-theoretic perspective* on risk analysis. The remainder of the paper explains it in more detail and discusses how it can be applied in numerous risk analysis settings.

Understanding the mathematical details of sheaf theory requires background in abstract mathematics, especially algebraic topology and category theory. This paper does not require that background and does not seek to impart such a detailed technical understanding. For that, we recommend existing excellent textbooks and monographs such as the one by Rosiak (2022). Rather, the goal of this paper is to explain the sheaf-theoretic perspective in enough detail to be useful to risk analysts thinking about part-whole relationships—whether in complex sociotechnical systems, in risk perception and communication, or in risk management decision-making, governance, optimization and deliberation—so that readers can decide whether to learn more about it, and can perhaps apply the perspective even if they decide not to learn more about the details. We believe that the main ideas are useful enough, and accessible enough, to be explained with useful clarity (but not complete precision) for risk analysis applications without requiring mastering the mathematical framework and abstractions needed for a full understanding of sheaf theory. Our central aim is to articulate the sheaf-theoretic perspective in risk science: The recognition that much reasoning about risk and risk management is naturally fragmented, and that analyzing how fragments relate (via restriction maps, discussed in the next section) can help diagnose where and why fragments do not align and when reconciliation and synthesis is possible. We think this perspective enables clearer communication, more consistent integration of evidence, and stronger decision support in risk analysis even without the full technical formalism of sheaf theory. This paper, therefore, is intended as an introduction and invitation to learn about a perspective based on a unifying mathematical theory that we believe is promising for shedding new light on many risk analysis topics, while acknowledging that full mastery of sheaf theory requires additional depth and effort beyond what is offered in this paper—as discussed in the references.

Our expository approach is, therefore, that, although key mathematical insights can only be expressed with full rigor, precision, generality, and elegance using the specialized language of category theory, they can be usefully approximated in ordinary language. That is what we seek to do. For example, a "sheaf" might be described in ordinary language as a mathematical system that assigns to each region of a space a set of allowable descriptions (called "sections") for that region—such as data, models, or properties—in such a way that (i) each section can be restricted consistently to smaller regions, and (ii) whenever locally consistent sections are specified on overlapping regions, there is one and only one way to glue them together into a section over the whole region. Such verbal explanation is suggestive but not rigorous. It does not specify exactly what is meant by a "space" or a "region" or "descriptions" or "gluing" local descriptions together. The following sections explain these ideas a little further, but our main approach is to rely on such verbal exposition to communicate the main ideas, albeit only imperfectly, so that it can be seen how they can be applied in risk analysis. To achieve a more precise and general understanding, we invite interested readers to learn more from Rosiak (2022), Ayzenberg et al. (2025), and other cited references. For readers who prefer a slightly more precise exposition of the sheaf-theoretic perspective without engaging with the cited references, we also provide some explanations and examples that require only linear algebra, as a compromise between merely verbal explanations and full technical explanations.

Beyond introducing the sheaf-theoretic perspective conceptually, this paper and its supporting information also seek to suggest some of its practical utility for risk science through a sequence of diverse applications in risk analysis, decision analysis, systems reliability, and related fields. These examples, drawn from causal inference, exposure–response modeling, engineering reliability, social risk perception, and multi-agent risk governance, show how the sheaf-theoretic perspective helps identify structural misalignments in fragmented information and clarify when coherent integration is possible. Although most of these topics were originally analyzed by other methods, we discuss how sheaf-theoretic reasoning might be applied to diagnose and sometimes resolve persistent problems in risk science. Each example can be viewed as showing how fragmented information can be mapped, assessed for coherence, and integrated into more robust decision support.

Misinformation and disinformation provide especially vivid examples of this broader challenge. If all agents shared the same information as common knowledge (in the sense of Aumann 1976), then systematic misrepresentation or deception based on hidden private information would be impossible, although errors, or misspeaking in communication, might still occur. But in real-world systems, siloed knowledge, semantic ambiguity, selective disclosure, and strategic distortion create the conditions for conflicting messages and false beliefs (Aven and Thekdi 2022; Jamalzadeh et al. 2022). Sheaf theory provides a useful perspective for diagnosing and addressing these problems by asking whether and under what conditions the information held or reported by different agents can be glued together into a consistent whole.

The rest of the paper presents sheaf theory as both a diagnostic lens for detecting incoherence and a design toolkit for improving integration in risk systems. We begin with foundational examples at the level of individual decision-making and proceed to challenges in teams, organizations, and society-wide coordination—particularly under uncertainty, incomplete trust, and imperfect communication. Misinformation and disinformation are treated as salient examples, but the paper's central focus is on understanding when and how fragmented inputs can be meaningfully combined, and when they cannot.

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Kay & King (2000) https://wwnorton.com/books/9781324004776 might quibble about the scale, boundedness, determinism of the "grand-world" definition presented here.

1.1 | Motivating Examples and Applications

In 1981, psychologists Amos Tversky and future Nobel laureate Daniel Kahneman posed the following pair of choices, to be considered concurrently (Tversky and Kahneman 1981):

Gains Choice

A. Gain $240 for sure

B. 25% chance to gain $1000, 75% chance to gain $0

Losses Choice

C. Sure loss of $750

D. 75% chance to lose $1000, 25% chance to lose $0

Each respondent had to choose both one member of the pair (A, B) and one member of the pair (C, D). Thus, the four logically possible combined choices are A&C, A&D, B&C, and B&D. They found that most people (73% of respondents) selected A&D—preferring the certainty of $240 (A) over the risky prospect (B) with a slightly higher expected monetary value (EMV = $0.25 \times \$1000 + 0.75 \times \$0 = \$250$), and also preferring the risky loss (D, EMV = –$750) over the sure loss (C). Thus, most respondents were risk-averse for gains but risk-seeking for losses—a well-known pattern in behavioral economics and in the descriptive theory of choice that Tversky and Kahneman pioneered, Prospect Theory (Tversky and Kahneman 1981).

Only 3% of respondents selected the least popular pair, B&C. Yet, as Tversky and Kahneman noted, "*the combination of B and C is definitely superior to the combination A and D*." That is, the least popular choice, B&C, unambiguously dominates the most popular, from the standpoint of rational choice theory, according to which people should, when possible, choose alternatives that make their preferred outcomes more likely. Specifically, B&C yields a 25% chance of gaining $250 (i.e., $1000 gain—$750 loss), and a 75% chance of losing $750. In contrast, A&D yields a 25% chance of gaining only $240, and a 75% chance of losing $760. All respondents preferred the former distribution (25% chance to win $250 & 75% to lose $750) over the latter (25% to win $240 & 75% to lose $760) when the question was posed in those terms, but most still selected A&D when choices (A or B) and (C or D) were presented separately.

This experiment demonstrates a fundamental challenge for decision-making under risk: Most people do not consider all the choices they must make and assess the probabilities of outcomes from different combinations of choices, as normative principles of decision and risk analysis would prescribe. Instead, they view each choice as its own "small world" (Savage 1954) —a simplified, well-defined context with its own known sets of choices, possible outcomes, and conditional probabilities of outcomes given choices—within which standard rational models such as expected utility theory apply and decisions can be optimized (e.g., by maximizing expected utility). Each small-world choice is viewed in isolation without considering other concurrent choices or the "grand world" view in which the decision-maker receives the joint outcomes from all of the individual choices (Qiu and Ahn 2021; Savage 1954). This *narrow framing* of each choice in isolation can—and in practice predictably does—lead to sub-optimal choices. Another way to view this problem is as a failure of compositionality: Isolated choices that do not violate rationality constraints (such as being undominated) cannot necessarily be combined or composed into a combined choice that satisfies the constraints.

Jackson (2019) https://www.wiley.com/en-us/Critical+Systems+Thinking+and+the+Management+of+Complexity-p-9781119118398, Kay & King (2000) https://wwnorton.com/books/9781324004776 question whether component sets can even be deterministically isolated from their environments.

Other difficulties in both individual and group decision-making can be viewed similarly in terms of tensions between what seems to make sense (i.e., satisfies desired logical or normative constraints) for each part separately and what makes sense for the whole consisting of all of the parts. Section 3 presents numerous examples of important challenges in individual, team, organizational, collective, and distributed choice theory that can be understood in terms of this tension. More generally, contemporary risk analysis of complex sociotechnical systems increasingly confronts systems whose behavior cannot be fully understood by examining individual components in isolation. Local observations, beliefs, preferences, actions, and communications among multiple people and software systems may interact in unexpected ways. This potential for unanticipated interactions is magnified when local actors include AI-enabled agents, machine learning models, or automated control systems whose internal logic may not be transparent to human overseers or to each other. As such systems make decisions based on partial, locally available information, the challenge of determining whether their collective behavior will align with broader human goals for the systems becomes both more urgent and more difficult.

This understates the problem. AI, machine-learning systems are statistically trained on "partially, locally available information" *from the past!*

Table 1 presents several examples of this challenge as it arises in diverse risk analysis applications. The table uses the special terms "local section" (also called simply a "section") to refer to local information and data and "global section" to refer to a whole set of coherent information to which the local sections contribute if they can indeed be joined together coherently. These are terms from the mathematical discipline of sheaf theory, as explained in the next section. They are defined rigorously, albeit abstractly, in mathematical sheaf theory, but we believe that they also have considerable intuitive appeal and informal heuristic value for risk analysts thinking about how complex sociotechnical systems can fail due to misalignment of the partial information in their local sections, that is, when it is not possible to assemble or compose them (or "glue" them, to use another sheaf-theoretic term with a precise mathematical definition and a strong intuitive counterpart) into a coherent whole.

Whether modeling cascading failures in power grids, tracing the spread of misinformation during public health emergencies, coordinating decentralized pandemic responses, or assessing resilience in interdependent infrastructure, risk analysts must grapple with how local interactions among system components and subsystems shape global outcome probabilities. In doing so, they must confront the reality that human and software agents often must make decisions with fragmented, incomplete,

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TABLE 1 | Illustrative risk analysis applications of sheaf-theoretic concepts.

Application	Local information ("Local sections")	Global coordination ("Global section")	Challenge	Possible risk management implications
Emergency response coordination	Each participant's partial situational awareness	Full situational picture enabling coordinated actions	No global section ⇒ Misallocation, delayed rescue	Build communication hubs or synchronization protocols to create coordination points.
Cyber security threat detection	Anomaly reports at different nodes	Integrated threat recognition across the network	No global section ⇒ Attack goes undetected	Design information sharing and aggregation mechanisms to stitch local anomaly reports into global threat detection.
Power grid risk management	Local sensor readings (voltage, frequency)	System-wide stability assessment to prevent blackouts	No global section ⇒ Local fixes worsen global instability	Deploy wide-area monitoring and control systems to create common situational awareness across substations.
Risk communication in public health	Agency-specific data and messaging	Consistent public health guidance	No global section ⇒ Confusion, mistrust, policy failure	Centralize or coordinate messaging channels to ensure that fragmented agency views are reconciled into unified public guidance.
Detecting an outbreak (pandemic or foodborne illness)	Early reports from clinics, hospitals, labs	Recognition of a significant outbreak trend early	No global section ⇒ Outbreak goes unrecognized until widespread	Create integrated early-warning surveillance systems that can combine scattered initial reports into a coherent outbreak signal.
Recognizing industrial process safety degradation	Incident and near-miss reports from different units	Early detection of systemic risk growth	No global section ⇒ Escalating risk without early intervention	Implement cross-unit safety data aggregation and joint risk review protocols before critical thresholds are crossed.
Recognizing shifts in urban patterns (crime, illness, suicide)	Local incident reports from neighborhoods, clinics, police	Citywide or region-wide pattern recognition suggesting intervention	No global section ⇒ Missed opportunity for preventive action	Build multi-agency information fusion centers and analytics to detect emerging trends that individual units cannot see.

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and sometimes conflicting or deliberately distorted knowledge. Traditional risk modeling tools—such as probabilistic graphical models, causal Bayesian networks, influence diagrams, and simulation-based approaches—are powerful for representing explicit views of how the world works and for making probabilistic calculations, but they do not emphasize representing or resolving inconsistencies among multiple perspectives, data sources, or agents. Both deliberate disinformation-induced disruptions to infrastructure coordination (Jamalzadeh et al. 2022) and lack of coordination and consistency in responding to crises from COVID-19 (Trump and Linkov 2022) to wildfire evacuations highlight the practical need for methods to assess when global integration has been compromised, and this requires new risk modeling methods.

The "system-of-systems" (SoS) perspective (Haimes 2019) provides a useful conceptual framework for addressing many of the challenges of modeling and managing risk in complex sociotechnical systems. The approach presented here builds in part on the SoS view's understanding that independently governed subsystems—each with its own goals, models, and decision processes—interact to produce emergent, system-wide behaviors. This view has proven especially useful in fields such as infrastructure resilience, emergency management, and risk governance (Haimes 2019). However, while SoS models capture the structural complexity and distributed control typical of real-world risk systems, they often lack a unifying mathematical framework for diagnosing whether the local models, policies, and information can actually fit together into a coherent whole. Moreover, they rarely offer tools for systematically reconciling mismatched assumptions, reasoning across multiple levels of abstraction, or detecting emergent inconsistencies—either from deliberate mis/disinformation or from errors and incompleteness of local views—in fused data, beliefs, or decisions. We believe that sheaf theory provides new ideas and methods that may help to fill these gaps.

The following sections explain how sheaf-theoretic ideas and tools can be used to model and improve various aspects of health, safety, and environmental (HS&E) risk analysis in real-world settings that exhibit structural features particularly suited to sheaf-theoretic analysis—especially distributed and partially incompatible sources of knowledge and control—such as partial and overlapping jurisdictional responsibilities; heterogeneous and dynamic data sources; adaptive, social, and adversarial behaviors; and distributed risk communication and decision-making under uncertainty.

Our applications of sheaf theory seek to augment existing tools in risk science by providing a common language and useful concepts and computational methods for encoding and analyzing the structural properties of information and observations (data); messages (truthful or not); beliefs (accurate or not); preferences (coherent or not); and actions across levels and actors in complex systems. We believe that sheaf theory suggests useful new ways to understand and model resilience, emergent behavior, and systemic fragility or robustness, helping risk analysts to move from intuitive system mapping to testable, interpretable models of real-world complexity, even in the presence of realistically imperfect and untrustworthy information.

2 | Sheaf Theory: Core Concepts in Brief for Risk Analysts

Sheaf theory formalizes the intuitive challenge familiar to every risk analyst: Given disparate, partial, and sometimes inconsistent observations from across a system, can we form a coherent overall picture? If not, where and why does consistency break down? While sheaf theory originated in mathematics (especially algebraic geometry and topology), it has found recent practical applications in sensor networks and data fusion, social systems, neuroscience, engineering, and increasingly, risk modeling. This section outlines the key concepts and terminology that risk analysts need to understand and apply sheaf-based tools in practical settings.

2.1 | What Is a Sheaf? (The Intuitive Picture)

Consider a distributed system in which multiple agents or regions have partial views or responsibilities—such as a network of air quality monitors, a patchwork of regional hazard assessments, or separate emergency response plans spread across government agencies. Each unit has access to local information, and perhaps some overlap with other units in what they observe or control. A key question addressed by sheaf theory is whether this distributed knowledge can be integrated into a single, internally consistent picture of reality or risk.

The following terminology is used to discuss part-whole relationships in sheaf theory. A *sheaf* is a mathematical structure, often represented in computational settings as a data structure that encodes three key components:

1. The data that can be assigned to each part or component of a system, captured through *sections* over those parts;
2. the rules for how data from different parts must agree on overlaps, described by *restriction maps*; and
3. the condition for whether all such local data assignments can be consistently combined—or *glued*—into a coherent system-wide assignment, called a *global section*.

More formally, a sheaf assigns to each "part" of a system—such as a node in a network or a spatial region—a set (or more generally, a mathematical object like a vector space or function space) of allowable *sections* over that part. These sections represent structured local information, such as numbers, functions, probability distributions, or decision rules. The collection of all sections over progressively smaller neighborhoods around a point determines the *stalk* at that point—a structure that captures all the local behaviors or data values that could occur near that location. In applied settings, a section over a part might be a real number (e.g., a temperature reading in a sensor network), a time series, or a probability estimate. In public health, it could be a risk estimate, a binary decision (e.g., close schools: yes/no), or a policy recommendation. The sheaf includes *restriction maps* that define how data over a larger part of the system must reduce or "restrict" to data over a smaller or overlapping part. These consistency requirements ensure that local pieces of information can, in principle, be checked for compatibility and integrated into a system-wide view.

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TABLE 2 | Sheaf-theoretic terminology and risk analysis interpretations of key concepts and terms.

Sheaf-theoretic term	Risk analysis analogy
Base space / parts / patches	The components of the system (e.g., regions, sensors, decision units)
Stalk	The set of all possible local data assignments at a point, based on nearby information; represents the space of potential beliefs or observations valid near that location
Section	A structured, internally consistent assignment of values or beliefs over a part of the system that obeys compatibility rules on overlaps
Restriction map	A rule for translating or reducing data from a larger part of the system to a smaller or overlapping one; for example, how a regional policy applies to sub-regions
Global section	A complete and consistent assignment of data to all parts of the system—a coherent "big picture," if one exists
Cohomology	A mathematical tool for identifying conflicts, gaps, or mismatches in how parts fit together—it explains *why* a globally consistent view may not be possible
Sheaf Laplacian	A generalized diffusion operator that models how information, risks, or beliefs spread through a system while respecting local compatibility constraints

For example, imagine two air quality sensors that monitor overlapping regions. One sensor measures temperature and humidity, while the other measures only temperature. To be consistent, both sensors should report the same temperature readings where they overlap—provided they measure temperature the same way. If one reports in Fahrenheit and the other in Celsius, then their readings must be converted to a common scale before they can be compared. In this case, the restriction map includes a rescaling step. More generally, restriction maps may be simple projections (e.g., selecting shared variables), unit conversions, or more complex rules for reconciling data, depending on the application. Together, these structures allow us to pose and answer rigorous questions about consistency, completeness, and coherence of local and global information in distributed systems.

Table 2 summarizes these key terms—the jargon of the field. It also includes two additional technical terms, cohomology and sheaf Laplacian, that will be discussed later. Table 3 provides some practical examples of how the basic concepts of sheaf theory can be applied to situations of interest in risk analysis. As a simple example to illustrate these concepts and terminology, imagine three regional air quality monitors, each responsible for tracking pollution in its zone. Monitor A covers region A, Monitor B covers region B, and Monitor C covers region C. The zones partially overlap: A and B both measure pollution near their shared border, as do B and C. The data each monitor reports for its region is called a *section* over that region. The shared boundary regions provide overlapping data, and the *restriction maps* describe how readings from one region should match or relate to readings from neighboring regions in those overlaps. If all three monitors produce data that agree where their regions overlap—and if those agreements fit together consistently across the whole system—then the system has *a global section*, representing a coherent, integrated view of air quality. If not, the sheaf structure helps identify where and how local reports fail to align. The language of sheaf theory provides a compact, flexible vocabulary for describing how parts of a risk system relate and whether they work together coherently.

2.2 | Why Should Risk Analysts Care?

Risk systems routinely involve fragmented, overlapping, or partially incompatible information. Sheaf theory provides a principled way to determine when such information can be combined into a coherent global understanding and where structural inconsistencies prevent it. Applications range from integrating heterogeneous sensor or model outputs, to coordinating emergency-response protocols, to diagnosing why expert groups or agencies assessing the same hazard reach different conclusions.

Sheaf methods also link analyses across scales, ensuring that local data, subsystem models, and system-wide summaries remain mutually consistent. They complement familiar tools such as fault trees, Bayesian networks, and influence diagrams by focusing on coherence rather than probability alone. When coherence fails, sheaf-theoretic analysis identifies the specific overlaps, assumptions, or interactions responsible. The examples that follow show how these ideas clarify real-world failures of aggregation, coordination, and interpretation in complex socio-technical and regulatory systems.

Section 3 introduces and illustrates key mathematical concepts and tools of sheaf theory; Section 4 and the supporting information apply them to many situations in which fragmented information challenges effective risk analysis; and Section 5 discusses broader implications for the field of risk analysis.

3 | Mathematical Tools for Sheaf-Theoretic Risk Analysis

Although it is rooted in abstract mathematics, sheaf theory offers concrete tools for solving real-world risk analysis problems. It provides practical computational and mathematical methods that can be applied to risk modeling, assessment, and management, giving analysts a principled way to reason about fragmented information, local constraints, and their global implications—

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TABLE 3 | Examples of sheaf-theoretic concepts applied to Risk Analysis.

Application domain	Base Units / Parts	Sections (local data assignments)	Restriction maps (compatibility across overlaps)	Global section
Air quality monitoring	Regional sensors or agencies	Local pollution measurements	Agreement on values at shared boundaries	Consistent map of air quality across regions
Early-warning systems	Sensors or monitoring units	Locally inferred hazard signals	Alignment of warning thresholds and triggers	Unified system-wide alert or escalation state
Public health messaging	Local, state, and federal agencies	Health guidance or mandates	Consistency of advice in overlapping jurisdictions	Clear, aligned communication to the public
Evacuation planning	Counties or jurisdictions	Local evacuation plans and routes	Compatibility of protocols at shared borders	Integrated, conflict-free evacuation strategy
Supply chain risk assessment	Manufacturers, suppliers, shippers	Local risk assessments (e.g., delays, failures)	Shared risks and handoffs at supply interfaces	Holistic supply chain risk profile
Systems reliability modeling	Subsystems or components	Local failure or reliability probabilities	Propagation via system logic (e.g., series/parallel)	Overall system reliability estimate
Cyber security (Bayesian networks)	Hosts, nodes, or subnetworks	Breach probabilities or attack likelihoods	Dependencies from attack graph or topology	Coherent overall breach risk assessment
Multiscale biological models	Molecules, cells, tissues, individuals	Gene/protein states, cellular responses, symptoms	Compatibility of models across biological levels	Biologically plausible causal model (e.g., exposure to disease)
Food safety & QMRA	Food chain steps or regions	Contamination levels, dose-response, growth models	Consistency of assumptions across steps/agencies	Quantitative risk estimate from source to outcome
HAZMAT routing	Road segments or regions	Local risk assessments and routing constraints	Compatibility of routing policies at boundaries	Feasible, legal end-to-end HAZMAT route
Power grid resilience	Substations, lines, generators	Load, capacity, and state data	Electrical coupling rules between nodes	Stable power distribution under constraints
Risk communication & SARF	Social groups, media spheres, institutions	Beliefs or trust levels about health risks	Agreement or influence across social overlaps	Coherent public risk perception and trust
Collective choice (Mitigation)	Voting districts or stakeholder groups	Preferences over mitigation options	Negotiation or aggregation over shared interests	Agreed-upon collective decision (e.g., funding levees)
Global risk governance	Nations or international bodies	Local threat models, policies, and targets	Compatibility of treaties, goals, or technical protocols	Coherent multinational strategy or governance framework

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particularly in distributed systems with overlapping data sources, policies, or responsibilities.

This section expands on Table 2's summary of the key concepts and terminology of sheaf theory. It introduces foundational mathematical concepts and computational tools from applied sheaf theory and shows how each supports practical reasoning about distributed, uncertain, or fragmented systems. Parts of this section assume familiarity with key concepts of linear algebra (including bases, kernels or null spaces, and column spaces). Readers uninterested in this technical background may simply read the examples.

3.1 | Technique 1: Gluing Together Global Sections—Testing for System-Wide Consistency

When data about a system or situation comes from multiple sources—as is common in applications such as environmental or medical monitoring, policy integration, distributed emergency planning, or tracking of risk communications on social media—analysts typically seek to combine these multiple local inputs into a single coherent overall picture of the environmental, biological, social, or organizational conditions of interest. In sheaf theory, this process of combining local information into a coherent global picture is called *gluing*, and it succeeds only when the local data agree consistently on all overlaps. The mathematical theory ensures that, when consistent local data exist, a unique global assignment can be constructed from them (Cakir et al. 2020). If the sources disagree or are only partially compatible, it may be unclear whether a coherent global interpretation is possible. Sheaf theory offers a formal method for answering this question. Intuitively, it provides methods to check whether local data and constraints (e.g., that flows of traffic or energy or materials must be conserved) can be combined, or "glued together," into a consistent global assignment (Cakir et al. 2020). When such an assignment exists, it is called a *global section*. This is one of the most useful and actionable ideas in applied sheaf theory.

A global section is a complete assignment of values across an entire system that satisfies all the constraints that apply locally. If such a system-wide solution exists, it represents a fully coherent model,, meaning one in which all local observations, measurements, or assumptions can be jointly satisfied without contradiction. Its existence tells us that the pieces of the system are structurally compatible. Its absence signals a deeper conflict—one that cannot be resolved simply by filling in missing values or adjusting inputs. Identifying whether a global section exists is often the first step in determining whether fragmented, overlapping, or partially observed data can be meaningfully integrated into a unified understanding of risk.

Table 4 outlines the work flow for using data to test for global sections. To apply this technique in practice, the analyst begins by breaking the system into distinct parts or *patches* (more formally, open sets, or parts of a base space). These are local components that each provide a partial view of the whole.

The patches correspond to domains over which local information is available. These might be different air quality sensors, policy jurisdictions, model components, or time slices. The next step is to identify *overlaps* between patches—places where data or rules from different components must agree or be reconciled. For instance, two air pollution sensors monitoring the same neighborhood, or two emergency response plans covering adjacent counties should give consistent results where their information overlaps.

The core technical step involves assembling a matrix—called the *coboundary matrix*, denoted D^0 —that records how the system's patches relate to one another through their overlaps. In simple applications, D° functions as a signed incidence matrix: Each row represents an overlap between two patches, and each column corresponds to a patch. The entries are typically 1, −1, or 0, depending on how each patch contributes to that overlap. A row with a +1 in one patch's column and a −1 in another's column expresses the direction in which the difference across that overlap is taken (all other entries being 0). The observed values or constraints on overlapping regions are collected in a vector ω, one entry per overlap. The gluing condition is expressed algebraically as $D^0x = \omega$: a global assignment x exists exactly when the proposed patch-level values reproduce all observed overlap values. In the common case where overlaps are pairwise, rows contain exactly two non-zero entries. More general systems, in which three or more patches jointly constrain a shared region, produce rows with more non-zero entries.

In real-world settings, especially those involving sensor noise or estimation error, an exact solution to $D^0x = \omega$ may not exist. In such cases, a least-squares approximation identifies the global assignment x that comes closest to explaining the observed overlap values, and the residual $||D^0x-\omega||^2$ quantifies the remaining inconsistency. The "consistency radius"—the smallest perturbation to the observed values ω that yields an exact solution—provides a useful measure of how nearly coherent the system is and highlights which overlaps contribute most to the misfit.

The linear formulation $D^0x = \omega$ captures the essential idea of sheaf-theoretic consistency and is sufficient for the examples developed here. More general sheaf models allow nonlinear or context-specific "restriction maps," time-varying relationships, or stochastic overlap data (Schmid 2025; Joslyn et al. 2020), but these generalizations do not change the basic insight: A system is structurally coherent when locally valid information can be aligned across overlaps to form a consistent global assignment. The linear case provides a clear and computationally tractable foundation for diagnosing when this is, or is not, possible.

3.1.1 | Example: Combining Air Pollution Measurements from Heterogeneous Sensors

Pham et al. (2022) model a distributed air pollution monitoring system that uses low-cost particulate matter (PM2.5) sensors. Their goal was to integrate data from multiple sources—including optical PM sensors and traffic cameras—to generate reliable, real-time pollution estimates in urban neighborhoods. In this system, each sensor—whether it measures particulate levels or vehicle counts—is treated as a patch. These patches overlap in space and function.: For example, multiple PM sensors may cover the same street corner, while a nearby camera records traffic patterns

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TABLE 4 | Workflow for testing for existence of a global section using sheaf theory.

Step	What You Do	What It Means
1. Identify patches	Break the system into local parts—e.g., sensors, agencies, time slices, or models.	Each patch corresponds to a domain where local data or policy information is available.
2. Identify overlaps	Determine where patches intersect—i.e., where two components share a location, population, or functional boundary.	These overlaps impose consistency constraints: Values from different sources should agree on shared domains.
3. Build a table of constraints, D^0	Create a matrix that encodes how patches are connected through their overlaps.	D^0 (the "coboundary matrix") expresses how local data assignments are related across overlaps
4. Assign observed values on overlaps (ω)	Compile a vector of observed data or policy values on the overlapping domains.	This "cochain" ω records the values that are actually assigned or observed on overlaps.
5. Solve $D^0x = \omega$	Try to find a vector x of local data assignments to patches that satisfies the observed constraints.	A solution x represents a coherent assignment—a global section. No solution implies structural or data inconsistency.
6. Interpret the result	Check whether a consistent solution exists and, if not, analyze where it fails.	Identifies misalignments in measurement, modeling, or policy assumptions across components.

that influence local emissions. The key challenge is to determine whether these data streams can be combined into a consistent global model of air quality.

Using the sheaf-theoretic framework, the authors constructed a coboundary matrix representing the relationships among sensors and overlaps. The cochain ω was assembled from observed discrepancies between sensor readings and inferred pollution levels based on traffic data. Importantly, the method is *causal* in the sense that vehicle counts are treated as upstream drivers of local pollution surges. These counts are translated into expected PM2.5 contributions using empirical emission models and are then used to adjust or correct the raw readings of nearby sensors. Solving the equation $D^0x = \omega$ enabled the researchers to determine whether a consistent pollution map could be constructed—and if not, which sensors or assumptions were contributing to the conflict. They found that the method could successfully: (a) Automatically identify faulty or inconsistent sensors; (b) Filter out local discrepancies that conflicted with upstream traffic-derived pollution signals; and (c) Compute a *consistency radius*—a threshold indicating how much deviation is needed to reconcile the data.

The result was a "self-filtering" sensor network that gave more accurate and robust PM2.5 estimates than previous methods, especially in areas with noisy or unreliable measurements. Compared to naïve averaging of measurements, the sheaf-based method reduced the mean absolute percentage error by more than 12%, demonstrating both practical utility and diagnostic power. This example illustrates how the abstract concept of a global section can be translated into a concrete tool for data integration, error localization, and system coherence testing in real-world risk systems. It shows that solving $D^0x = \omega$ provides constructive techniques for assessing how well the components of a complex system can be reconciled—and for detecting and excluding faulty sensor readings or other anomalies that would otherwise prevent a consistent global interpretation of the data.

3.1.2 | Example: Integrating Heterogeneous Geolocation Data for Wildlife Tracking

A complementary example of global section analysis comes from a study by Joslyn et al. (2020) that used sheaf theory to integrate diverse geolocation data for monitoring wildlife, specifically black bears in the Appalachian region of North Carolina. The researchers faced a common risk-analytic challenge: Data from multiple observation systems—in this application, GPS collars, radio telemetry, and field notes—each provided only partial, noisy, or context-dependent glimpses of the bears' true locations and movements. Some of these data were in standardized formats (e.g., latitude–longitude coordinates), while others were qualitative or required expert interpretation (e.g., "bear sighted near the river bend"). The goal was to combine these disparate sources of information into a consistent spatiotemporal reconstruction of each animal's trajectory. To do this, the researchers treated each type of observation and each time point as a patch in a sheaf structure. Overlaps occurred where multiple observations referred to the same time and region—say, a GPS reading and a visual sighting recorded within minutes of each other. The coboundary matrix D^0 encoded which patches (observations) overlapped and how they constrained one another. The cochain ω captured the observed differences or uncertainties—such as spatial disagreements between sensors or textual ambiguities.

Solving the equation $D^0x = \omega$ yielded candidate location assignments x that would reconcile the conflicting and incomplete reports. Where no exact solution existed, least-squares approx-

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imations were used to estimate the best-fit trajectory and to measure inconsistency across the system. Notably, the sheaf framework did not require converting all observations into a common coordinate system beforehand—a significant advantage over many traditional data fusion techniques. This flexibility proved especially useful in integrating qualitative and probabilistic reports with quantitative sensor data. In direct comparison with more conventional methods for smoothing and predicting trajectories (a dynamic linear model estimated from data via a Kalman filter), the sheaf-based approach showed comparable performance in reconstructing bear paths but offered clearer diagnostics when observations could not be reconciled. In other words, while both methods could track the bears, only the sheaf model could say why some data sources disagreed, and where uncertainty was too high to justify a coherent interpretation. This example illustrates how sheaf-based integration of heterogeneous data ("data fusion") applies to data that differ not just by location or scale, but also by type and reliability.

3.1.3 | Example: Incompatible Risk Scales and Alert Levels Across Jurisdictions

For a non-numerical application, imagine three overlapping public health jurisdictions: County A, County B, and their shared urban center, Region AB. Each is responsible for issuing a COVID-19 risk alert to the public. While they aim to convey the same underlying phenomenon, public health risk, their internal systems for assessing, and reporting that risk differ. County A uses a numeric risk index, assigning values from 0 to 10 based on a quantitative model. County B uses a color-coded system, drawing on qualitative assessments. In contrast, Region AB—where the two jurisdictions overlap—adopts a standard ordinal scale for public messaging: {LOW, MODERATE, HIGH}.

In sheaf-theoretic terms, each region's possible internal representations form its *stalk*—the set of values it can assign to its domain. The stalks differ:

- County A's stalk: Numeric scores (e.g., $\{0, 1, 2, \ldots, 10\}$)
- County B's stalk: Symbolic color codes (e.g., {"Green", "Yellow", "Red"})
- Region AB's stalk: Shared alert levels {LOW, MODERATE, HIGH}

To coordinate messaging across jurisdictions, *restriction maps* are defined that translate each county's internal risk scale into the shared alert levels used by Region AB:

- County A's mapping: 0–2 → LOW, 3–5 → MODERATE, 6+ → HIGH
- County B's mapping: "Green" → LOW, "Yellow" → MODERATE, "Red" → HIGH

Now suppose County A reports a risk index of 6, while County B reports "Yellow." Applying their respective restriction maps, County A's index of 6 maps to HIGH; and County B's "Yellow" maps to MODERATE. Although both regions are internally consistent, no global section exists which is consistent with both local assessments when mapped to the shared overlap. The structural failure stems not from measurement variability or reporting error, but from lack of semantic coordination. The restriction maps do not allow the local data to agree on the overlap, and so gluing fails.

3.2 | Technique 2: Sheaf Cohomology—Diagnosing Inconsistency and Structural Obstruction

While the existence of a global section reveals that a system of local data and constraints can be reconciled and integrated (glued) into a coherent whole, the *non*-existence of a global section raises a useful question: *why not?* What exactly prevents the pieces from fitting together? The second major computational technique of sheaf theory, *cohomology* calculation for a sheaf, provides insights and answers. In particular, it helps to distinguish among three possible causes of misalignment of local data that might prevent it from assembling into a consistent whole:

- Noise or error in the data (which might be corrected),
- Missing information or insufficient overlap (which might be supplemented), and
- Structural obstructions inherent in how parts of the system are organized and connected (which may require redesign of its layout or operation).

Sheaf cohomology is especially valuable for identifying structural misalignments that cannot be resolved by simply improving or adding data. It detects higher-order incompatibilities that persist even when all pairwise overlaps agree, analogous to trying to fit tiles around a circle only to find they leave a gap.

In the linear setting, the cohomology calculation builds on the same ingredients used for gluing: The vector x of patch-level assignments, the first coboundary matrix D^0, and the residual vector $\omega = D^0 x$ that records mismatches on overlaps. A second coboundary matrix D^1 encodes how overlaps themselves are linked into cycles, such as when three patches form a triangle of pairwise overlaps. The condition $D^1\omega = 0$ formalizes the idea that these mismatches "close up" around every loop: There is no net discrepancy when we follow overlaps around a cycle. Sheaf cohomology asks two questions about ω: (i) does it pass this loop-consistency test ($\omega \in \ker(D^1)$) and (ii) if so, can it nonetheless fail to come from any global assignment x ($\omega \notin \mathrm{im}(D^0)$)? The answers are summarized by the first cohomology group H^1, discussed next.

The relationship between local and global consistency can also be summarized using the first cohomology group H^1. As described above, the matrix D^1 checks whether the mismatch pattern ω around each loop "closes up"—that is, whether the signed differences recorded on overlapping regions cancel when followed around every cycle of patches. All ω that satisfy this loop-consistency condition lie in $\ker(D^1)$: they exhibit no contradictions when checked cycle by cycle. Among these, some ω vectors actually arise from a global assignment x, meaning $\omega = D^0 x$ for some choice of patch-level values; these lie in $\mathrm{im}(D^0)$. The first cohomology group $H^1 = \ker(D^1)/\mathrm{im}(D^0)$ captures the remainder: Mismatch patterns that are locally and loop-wise consistent but still cannot be explained by any

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global assignment. In linear-algebra terms, its dimension satisfies $\dim(H^1) = \dim(\ker(D^1)) - \dim(\mathrm{im}(D^0))$, which counts how many independent directions of unresolved inconsistency remain after removing all parts that could be generated by coherent global data. These residual patterns indicate structural obstructions, that is, features of the system's pattern of overlaps and connections that prevent complete reconciliation of local information even when all pairwise and cyclic constraints agree. If $\dim(H^l) = 0$ ("trivial cohomology class"), then every locally and loop-wise consistent pattern can be produced by some global assignment, and there are no hidden contradictions—no cases in which local consistency fails to extend to global consistency. If $\dim(H^l) > 0$ ("nontrivial cohomology class"), then one or more structural obstructions are present—features of how the system's information is organized that prevent full global coherence despite local consistency.

3.2.1 | Sheaf Cohomology Beyond Linear Algebra

The linear-algebraic machinery built from D^0 and D^1 is only one instance of the broader framework of sheaf cohomology. The same principles apply when local data are not numeric vectors but logical propositions, regulatory requirements, probability distributions, expert assessments, or other structured objects. What persists across settings is the basic architecture of a sheaf: A collection of local data (stalks), defined over regions or agents (the base space), together with restriction maps that specify what it means for assignments to agree on overlaps.

For example, when agencies hold different sets of propositions or policy assumptions, stalks can contain logical statements or policy alternatives, and restriction maps preserve entailment or consistency across shared domains. Cohomology then tests whether a single globally consistent set of conclusions exists; a nontrivial H^l reveals structural contradictions that cannot be resolved by adjusting individual propositions alone. Similar issues arise when regulatory requirements differ across jurisdictions, or when local policies impose incompatible mandates on shared populations or resources. In probabilistic modeling, stalks may consist of local probability distributions, and restriction maps implement marginalization or conditioning. A global section corresponds to a coherent joint distribution compatible with all local marginals. Nontrivial cohomology indicates that no such joint distribution exists—an issue familiar in distributed sensor fusion, expert elicitation, and Bayesian networks when locally valid beliefs cannot be reconciled.

Although matrices D^0 and D^1 are not defined for these more general structures, abstract cohomology (via standard categorical constructions) can still be applied, with H^l measuring the extent to which locally valid models, rules, or beliefs can coexist without contradiction—and, if not, what structural features prevent this. By examining which restriction maps or overlaps contribute to a nontrivial cohomology class, analysts can identify where gluing breaks down—whether from missing information, poorly aligned assumptions, or deeper structural incompatibilities. These diagnostics can suggest remedies such as adding communication links, harmonizing assumptions, or redesigning interfaces between overlapping jurisdictions or models.

3.2.2 | Example: Coordinating to Complete Tasks in Distributed Systems

In distributed systems, agents—whether humans, computers, or AI decision modules—often must complete a shared task by a deadline based only on their own information and without centralized control. Decisions must be made in finite time, and each agent must give the same output in all situations it cannot distinguish based on its own observations or messages; this requirement is known as *causal consistency*. Examples include emergency teams deciding whether or how to evacuate, or autonomous systems coordinating traffic rerouting. But under what conditions is such decentralized coordination possible?

Sheaf-theoretic modeling provides a precise answer. Felber et al. (2025) show that a distributed task can be accomplished in finite time if and only if there exists a reachable moment in the system's evolution—called a *cut*—at which every agent's view (its locally observable history) contains enough information to produce a causally consistent output, and those outputs collectively satisfy the task specification. If no such cut exists, then no protocol, regardless of ingenuity or persistence, can succeed.

In this framework, the *task sheaf* encodes each agent's evolving local knowledge as a stalk, with overlaps representing shared observations, messages, or shared memory between agents. The stalks hold local knowledge (the agent's view), and the overlaps capture who shares what with whom. A *task* specifies which combinations of final outputs count as correct for any set of agent views, and a *protocol* is an algorithm each agent uses to update its local state and ultimately terminate with an output. A solution requires that every agent eventually terminates, produces a causally consistent output, and that these outputs jointly satisfy the task. Coordination succeeds only if, at some cut, the local views can be glued into a coherent global solution—that is, when the sheaf's restriction maps admit a global section.

Sheaf cohomology determines when such gluing is possible. If the first cohomology group H^l of the task sheaf is non-zero at every reachable cut, then no cut admits a global section, and the distributed task is unsolvable in finite time—not because of noisy data or poor protocols, but because of the system's structure. This insight applies directly to familiar risk-analysis problems:

- *Emergency response*: Teams may act on accurate but partial local views; without a moment when information aligns sufficiently, coordinated evacuation is impossible.
- *Public health surveillance*: Local offices may detect signals of an outbreak that never overlap enough to support a coherent regional alert.
- *Environmental regulation*: Agencies may rely on sound local models whose assumptions cannot be reconciled globally.

In such cases, cohomology not only diagnoses failure but suggests remedies: Additional communication links, harmonized assumptions, or shared metrics that make coordination feasible. This framework applies broadly to systems in which local actors must coordinate based on partial and evolving information, offering risk analysts a principled way to distinguish coordination

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problems solvable by improved protocols from those that require structural redesign.

3.3 | Technique 3: Restriction Maps—Modeling Compatibility Across Scales and Domains

In complex systems and organizations, information often flows among different scales, levels of a hierarchy, or domains—such as from national policies to local implementations, or from high-resolution sensor data to aggregated summaries. *Restriction maps* are a fundamental component of sheaf theory that formalize how information is translated consistently across these different contexts. In many risk analysis settings, the same phenomenon is described differently at different levels of resolution or across different domains. For example:

- A national air quality standard may be based on yearly averages, while a local ordinance responds to hourly pollution spikes.
- A federal health agency may classify diseases using one taxonomy, while a local health department uses another based on symptom clusters.
- Engineers monitoring structural integrity may work with detailed sensor data, while managers receive dashboard summaries or risk scores.

Ensuring that information is compatible across levels—and that summary-based decisions remain faithful to underlying data—is a core challenge in multi-scale risk analysis. Restriction maps address this by explicitly modeling how information is transferred and transformed across system components.

In simple terms, restriction maps describe how information in a larger or more comprehensive domain *restricts* to a smaller or more focused one. They formalize how a national policy is interpreted locally, how high-resolution data projects onto summaries, or how institutional knowledge filters or translates across organizational boundaries. More technically, restriction maps are the links between the "stalks" in a sheaf—the pieces of data or decisions associated with individual regions, agents, or domains. If the stalk over a large patch contains full knowledge (e.g., a high-dimensional risk model), the restriction map tells us how much of that knowledge is visible or usable in a smaller, overlapping patch (e.g., a field office with limited inputs). This matters whenever decisions depend on *incomplete views*. Restriction maps make explicit what is preserved, lost, or transformed as information moves between scales.

At their core, restriction maps define how data assigned to a larger region or context can be "restricted" or translated to a smaller one. For example, in a sheaf modeling air quality data, a restriction map might specify how a regional pollution measurement applies to a specific city within that region, helping to ensure that data remains consistent and meaningful as it moves between different parts of the system. In infectious disease surveillance, one system might classify influenza-like illness using symptom codes, while another uses diagnostic lab results. Restriction maps between these systems (conceptual or implemented in code) would show what information can be compared or aligned, helping analysts answer: *Do these systems measure the same thing? If not, where do they diverge?* Likewise, a risk model might output probabilistic forecasts or scenario trees. A public information officer may translate this into categorical warnings ("low," "moderate," "high"). Restriction maps in this setting formalize what is retained or lost in that translation. They can help analysts design warning systems that minimize distortion and preserve key distinctions.

Without restriction maps, it is easy to assume that different levels or actors are talking about the same thing, even when they are not. Miscommunication and misalignment often stem not from errors, but from mismatched representations of the same phenomenon. By modeling how information restricts and propagates, analysts can check whether local decisions respect upstream assumptions; detect when incompatible assumptions are being made across domains; and guide the design of better information summaries and data-integration methods. Moreover, as discussed later, restriction maps enable one of the most powerful capabilities in sheaf theory: *multiscale modeling*. They let us build systems where high-resolution models and coarse summaries coexist, interact, and stay logically consistent.

Restriction maps do more than enforce consistency across adjacent patches; they also underpin multiscale and hierarchical modeling. In many risk systems—such as nested public health agencies, layered regulatory frameworks, or multi-resolution sensor networks—information must flow coherently between higher-level summaries and lower-level observations or actions. Restriction maps make this flow explicit and testable. They define how a national guideline restricts to state policy, how detailed forecasts aggregate into regional risk indicators, or how city-level mitigation efforts relate to federal performance metrics.

Multiscale modeling is not only about passing information up and down a hierarchy. It is also about ensuring that decisions, interpretations, and models at different levels are mutually compatible and reinforce a coherent system-wide understanding. Sheaf theory gives risk analysts the tools to express and enforce this compatibility formally. It supports modular model design, in which components at different scales can be verified for gluing, and guides the construction of shared vocabularies and metrics that preserve meaning across levels. This makes restriction maps an essential foundation for modeling complex, multilevel systems where coherence across scale is both critical and fragile.

3.3.1 | Example: Restriction Maps in PM2.5 Sensor Fusion

As discussed earlier, Pham et al. (2022) present a sheaf-theoretic approach to fusing PM2.5 pollution measurements from a network of sensors. In their model, each sensor provides a local PM2.5 reading—treated as a stalk—and regional summaries (such as neighborhood averages) live on larger patches. Restriction maps specify how any proposed regional summary must agree with the fine-scale sensor readings it covers. They do not perform aggregation, but instead formalize what it means for coarse and fine data to be compatible. When these restriction maps fail to align—for example, when noisy or missing sensor data prevents any regional summary from restricting consis-

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tently to all its overlapping sensors—the sheaf structure reveals that no coherent global pollution map can be formed. In the stochastic setting considered by Pham et al., the consistency radius (the smallest perturbation to the observed overlap data that would allow perfect gluing) quantifies how nearly compatible the sensor network is and highlights where inconsistencies are concentrated. This illustrates how restriction maps govern data compatibility across scales, ensuring that high-resolution sensor data and regional summaries can be combined into meaningful alerts or policy-relevant assessments.

3.4 | Technique 4: Sheaf Laplacians—Modeling Constrained Diffusion and Signal Propagation

In many systems of interest to risk analysts, information, beliefs, risks, or failures do not stay confined to one place—they spread. An infectious disease crosses regional borders. A rumor or misinformation campaign ripples through online communities. A structural failure in one part of a complex engineered system causes cascading disruptions. Even trust, fear, or misperceptions about risk can diffuse through networks or populations of individuals or institutions. Understanding these patterns of spread is essential for identifying and mitigating systemic risks.

In familiar physical settings, flows are often modeled by using familiar techniques of applied mathematics to describe how quantities such as temperature, concentration, or pressure diffuse and average out among neighboring points. In continuous physical systems, such as the spread of heat through a metal plate, diffusion is governed by the *Laplacian operator* from differential equations. In networked systems such as electrical power grids, road networks, or social networks, a related idea is captured by the graph Laplacian, which is a matrix that encodes the connectivity structure of a graph and quantifies processes like diffusion, consensus formation, or signal propagation and smoothing on networks. However, classical Laplacians—whether continuous or graph-based—assume that every location in the system holds the same type of value (typically a number representing the diffusing quantity, such as temperature), and that the relationships among locations are symmetric and uniform. These modeling assumptions breaks down in many real-world risk systems, where local states are diverse, communication is filtered or constrained, and units or meanings differ across nodes. For example, a bridge sensor might report a stress tensor, while an adjacent sensor reports strain. A local government might have a risk assessment using different assumptions than a neighboring jurisdiction. A public health agency might hold a belief (probability distribution) over disease spread models, while a community influencer shares a simplified message on social media. In such cases, the underlying network is *heterogeneous*: Not all nodes use the same data types, representations, or communication rules.

Sheaf theory addresses the challenge of modeling interacting heterogeneous information by allowing each node or region to hold different types of data—typically (but not exclusively) represented by different vector spaces or logic structures. Restriction maps encode how they are related. A *sheaf Laplacian* generalizes the classical graph Laplacian to these more general settings (Hansen and Ghrist 2021). Instead of assuming that all nodes hold numbers (scalar values), it defines a matrix-like operator that represents the relationships and local data types defined by a sheaf. It can be used to model how information, beliefs, or influences diffuse—or fail to diffuse—subject to these constraints. Use cases in risk analysis include the following:

- For engineered systems such as power grids or transportation networks, failures or stresses may propagate—but not everywhere, and not equally. Certain paths may be blocked by policy, physical barriers, or detection systems. The sheaf Laplacian models these constraints explicitly. It helps simulate where risk is likely to move, and where it might pool or amplify.
- Social diffusion of beliefs or misinformation considers how risk perceptions about vaccine safety or wildfire evacuation evolve on social media. Users may share beliefs that are locally consistent (within their echo chambers) but globally contradictory. The sheaf Laplacian can model how such beliefs spread across communities with differing interpretations, highlighting where clarification or trust-building interventions (or, on the dark side, deliberately planted lies or mis/disinformation) can have the largest impact (Atkas et al. 2024).
- Multi-agency coordination and belief updating: These are agencies that coordinate on disaster response or risk forecasting and often update their beliefs based on incomplete or transformed information from others. The sheaf Laplacian allows analysts to model these filtered exchanges and study convergence or divergence of beliefs over time.

For the applied mathematical modeler, the sheaf Laplacian (and its eigenvalues) provides powerful insights into the dynamics of information spread in heterogeneous-information systems, including identifying (a) slow modes of propagation, which indicate where beliefs or influences are "sticky" and change slowly; (b) structural bottlenecks (regions that block or slow diffusion), where communication constraints prevent alignment or delay diffusion; and (c) persistent disagreement or cycles, where signals recirculate without resolution. Identifying these features can inform the design of more effective monitoring, communication, coordination strategies, and robust intervention plans in complex risk systems.

3.4.1 | Example: Opinion Dynamics in Social Networks

A striking illustration of the power of sheaf Laplacians comes from recent work by Hansen and Ghrist (2021), who introduce a framework called *discourse sheaves* to model the evolution of opinions in social networks. In this framework, each node (e.g., an individual or institution) holds a private opinion—represented as a vector in some space—and communicates a version of that opinion to others, potentially filtered, scaled, or distorted by restriction maps. These maps may model simplification, selective emphasis, or even misrepresentation of the individual's true private opinion. The resulting structure—a sheaf over the social graph—encodes both what each agent believes and how it expresses those beliefs to others.

To simulate how such expressed opinions influence one another over time, Hansen and Ghrist apply a sheaf Laplacian to

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model opinion dynamics as a system of differential equations in which individual opinions evolve over time in response to the opinions of others. Under these dynamics, opinion states quickly converge to a global section of the sheaf—that is, to a system-wide assignment of opinions which is consistent with all the local expression rules. This offers a powerful formalization of consensus (or persistent disagreement) in systems where communication may be strategic, constrained, or noisy. Sheaf Laplacians can encode not just numerical averaging, but structured communication, distortion, and influence—critical ingredients for understanding the spread of beliefs, warnings, and behaviors in modern information ecosystems.

For risk analysts, the sheaf theory framework offers several insights. It helps to explain why public consensus may be difficult to achieve, even if each individual is internally consistent, when communication channels distort, suppress, or fragment information. It allows simulation of how misinformation, polarization, or echo chambers arise structurally—not just from irrationality, but from how local beliefs are communicated and interpreted. It also supports analysis of possible interventions, such as modifying communication pathways or harmonizing restriction rules, to promote system-wide coherence in risk understanding.

3.4.2 | Example: Misinformation Injection and Partial Recovery in a Small Network

To see in detail how a sheaf Laplacian can be used to model the diffusion—and potential correction—of misinformation in a very simple, small linear algebra model, consider a network of four agents {A,B,C,D} connected by five symmetric, trust-weighted communication links. The trust weights serve as restriction maps between agent beliefs (acting as scalar linear restriction maps between 1-dimensional stalks. When all stalks are scalars and all restriction maps are multiplications by trust weights, the sheaf Laplacian reduces to a weighted graph Laplacian, making this example directly comparable to classical diffusion.) Figure 1 shows the network.

Each agent holds a scalar belief, initially given by the vector:

$$x_0 = [\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}]^{\mathrm{T}} = [1.5, 0.8, 0.0, 0.2]$$

The corresponding coboundary matrix D^0, derived from the network structure and trust weights, is:

$$D^0 = \begin{bmatrix} 0.9 & -0.9 & 0.0 & 0.0 \\ 0.0 & 0.7 & -0.7 & 0.0 \\ 0.0 & 0.0 & 0.8 & -0.8 \\ -0.6 & 0.0 & 0.0 & 0.6 \\ 0.5 & 0.0 & -0.5 & 0.0 \end{bmatrix}$$

Each row of D^0 corresponds to an edge. Each column of D^0 corresponds to a node. For an edge connecting nodes u and v, the corresponding row in D^0 contains:

- A positive value equal to the trust weight in the column for node u,

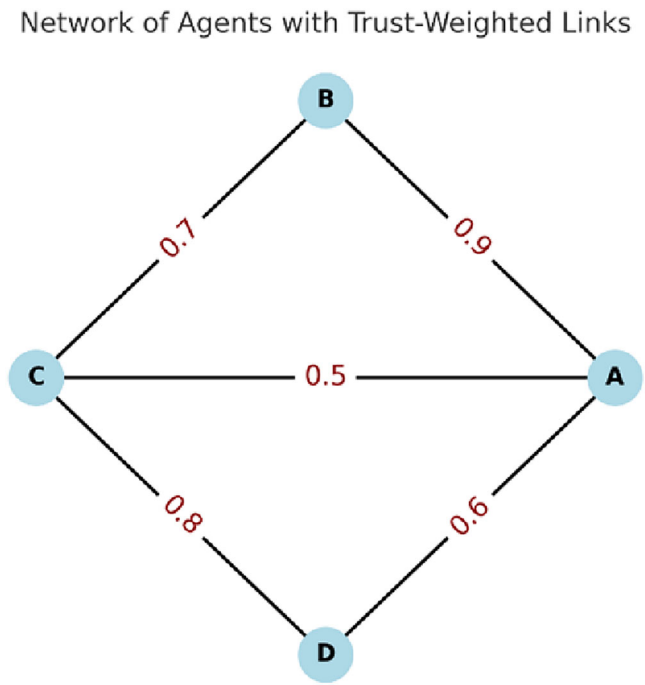

FIGURE 1 | A small example network.

- A negative value of the same magnitude in the column for node v,
- Zeros in all other columns.

This reflects how the belief at node u is "restricted" to the belief at node v, and vice versa, according to the strength of trust or influence along that edge. The choice of sign ensures that the matrix encodes directional difference between neighboring beliefs, consistent with conventional orientation (though the orientation is arbitrary and may be symmetrized). We assign each edge an arbitrary orientation (e.g., from left to right or alphabetically). Then each row of D^0 is built as follows:

- Row 1 (A–B): +0.9 in Column A, –0.9 in Column B
- Row 2 (B–C): +0.7 in B, –0.7 in C
- Row 3 (C–D): +0.8 in C, –0.8 in D
- Row 4 (D–A): +0.6 in D, –0.6 in A
- Row 5 (A–C): +0.5 in A, –0.5 in C

This matrix captures the weighted difference structure that governs belief interactions in the network. It is used to compute the sheaf Laplacian L, which governs diffusion dynamics.

The sheaf Laplacian $L = D^{0\mathrm{T}} D^0$ (where "T" denotes transpose) is then calculated via matrix multiplication, yielding:

$$L = \begin{bmatrix} 1.42 & -0.81 & -0.25 & -0.36 \\ -0.81 & 1.30 & -0.49 & 0.00 \\ -0.25 & -0.49 & 1.38 & -0.64 \\ -0.36 & 0.00 & -0.64 & 1.00 \end{bmatrix}$$

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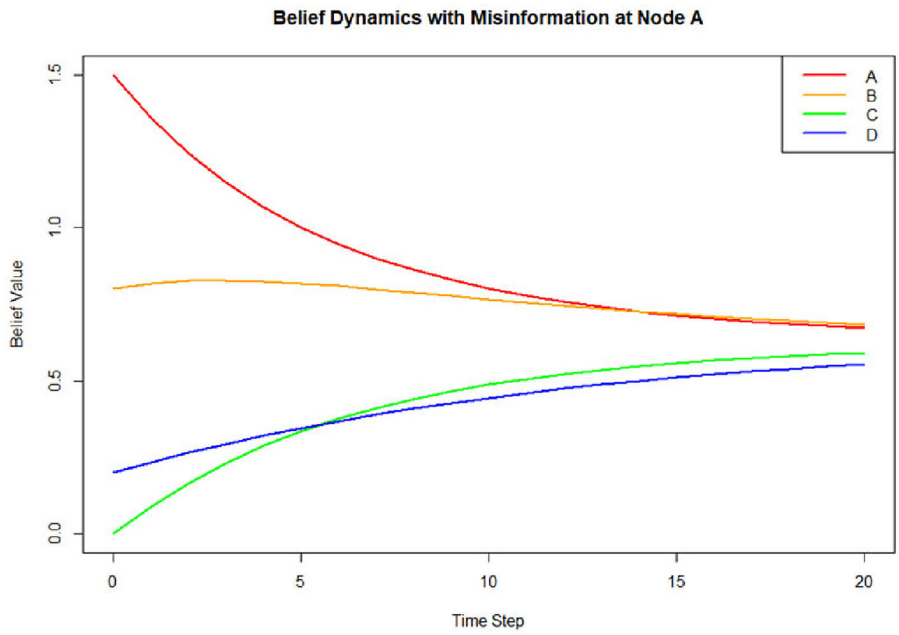

FIGURE 2 | Simulation results showing mutual adjustment of individual beliefs over time.
Source: The appendix provides the R code for Figure 2.

Simulating belief evolution over time using the discrete update rule

$$x_{t+1} = x_t - \eta \times L \times x_t, \text{with } \eta = 0.1$$

where x_t is the 4-component belief state vector t time t shows how misinformation introduced at a single node gradually diffuses through the network, influencing others before being corrected. Figure 2 shows the results of this simulation

The red curve shows the belief at node A, which starts at 1.5 due to an injected misinformation shock. Over time, A's belief steadily declines as it is influenced by its neighbors, particularly B and D, through trust-weighted averaging encoded in the sheaf Laplacian. However, A's inflated belief also causes noticeable upward drift in the beliefs of C and D (green and blue curves), which initially held much lower values. Node B (orange) experiences a more modest and transient elevation, due to its proximity to both A and C. As the update rule is iterated, all beliefs move toward convergence. But the final consensus reached across the network reflects a lasting influence of the initial misinformation: All nodes settle at higher beliefs than they would have in the absence of node A's initial misinformation. This illustrates how even a single localized distortion can systematically bias the eventual belief equilibrium—especially when it originates at a well-connected or trusted node. Importantly, the sheaf Laplacian framework enables both simulation and diagnosis of these effects. By analyzing how disagreement is structurally distributed (via the Laplacian's eigenvalues and residuals), one can identify which nodes or substructures are most vulnerable to persistent misinformation and where interventions might be most effective in restoring coherence. This highlights the potential value of sheaf-theoretic diffusion models for understanding, predicting, and mitigating the network-level impact of distorted or adversarial signals in risk communication systems.

While this simple expository example uses scalar-valued beliefs and simple linear diffusion, and could be analyzed using linear algebra (classical graph Laplacians) without sheafs, the sheaf Laplacian framework generalizes seamlessly to more complex and realistic settings. As emphasized previously, sheaf models allow each node to hold different types of data—including vectors, probability distributions, logic structures, or qualitative assertions—and permit flexible restriction maps that translate between them across edges. This flexibility is essential for modeling many realistic heterogeneous systems. Sheaf Laplacians extend the intuition of constrained diffusion to these richer, nonuniform systems, enabling risk analysts to track, predict, and mitigate belief misalignment even when the underlying data and communication rules are complex, diverse, structured, or domain-specific.

3.5 | Summary: Sheaf Theory as a Toolbox for Risk Analysts

This section has introduced four foundational analysis tools from applied sheaf theory—global sections, sheaf cohomology, restriction maps, and sheaf Laplacians—that offer practical support for reasoning about complex, distributed, or fragmented systems in risk analysis. While each technique builds on abstract mathematical structures, their relevance for risk analysts lies in how they help to address familiar problems of integration, misalignment, and coordination.

Global sections provide a test for whether diverse and locally valid data or decisions can be integrated into a coherent system-wide view. When such integration is not possible, sheaf cohomology reveals whether the failure is due to removable noise, insufficient overlap, or inherent structural obstructions that cannot be resolved without redesign. Restriction maps make explicit how

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data or decisions relate across overlapping contexts—modeling the translation of information across scales, jurisdictions, or domains, and identifying where semantic or definitional inconsistencies may prevent coherent communication or integration. Sheaf Laplacians extend these ideas to systems in motion, modeling how risks, beliefs, or failures propagate under structural constraints—such as through sensor networks, organizational hierarchies, or social media.

Together, these tools form a coherent analytical framework for diagnosing and addressing a wide range of decision-relevant problems in health, safety, environmental, and policy domains. They help analysts move beyond local rationality or internal consistency to ask whether a proposed set of observations, models, policies, or decisions can work together as a system. And when they cannot, they point precisely to the sources of misfit and the structural remedies needed to resolve them. The next section turns to applications. It reviews how failures of compositionality arise in real-world risk systems—ranging from individual beliefs and team coordination to regulatory policy and collective decision-making—and shows how sheaf-theoretic concepts and methods can help clarify, diagnose, and improve the design and operation of risk systems by enhancing the coherence with which their parts function together, thereby improving overall system performance.

4 | From Fragmented Parts to Coherent Wholes: Sheaf Theory in Decision and Risk Science

The preceding sketch of the main mathematical concepts of sheaf theory might give the impression that sheaf theory is a generalization of linear algebra that extends familiar ideas about solving systems of equations to more abstract and fragmented domains—and that it is, therefore, of more relevance to mathematicians than to risk analysts. This section argues that the core ideas and techniques of sheaf theory can shed useful light on much of risk science. Persistent, important, and fascinating challenges—from risk perception and communication to causal inference, exposure-response modeling, engineering reliability, collective risk management, and multi-agent governance—can usefully be viewed as instances of one underlying problem: How to reason and act coherently with fragmented information. As discussed in the previous sections, sheaf theory provides tools to diagnose when and how such integration is possible in well-specified systems. Even for social or sociotechnical systems that may not be well-specified, it offers fruitful metaphors for understanding and helping to resolve integration problems, suggesting how to improve coordination, detect structural misalignment, and enable better decision-making by identifying and resolving structural obstacles.

The supporting information presents a series of paradigmatic topics in decision and risk analysis with fragmented information. It seeks to show how the sheaf-theoretic perspective can help to illuminate structural failures or suggest corrective measures and changes in systems or organizational designs to improve coordination and decision-making in distributed risk assessment and risk management. The examples are organized along a conceptual progression from individual-level inconsistencies to coordination challenges within teams, to strategic and institutional interactions, to collective and societal risk management and risk governance, ending with emergent dynamics of risk perceptions in social networks. We believe that the sheaf-theoretic perspective has much to offer at each step.

Table 5 summarizes the key results via a series of challenges or impossibility theorems in its left column, paired with proposed resolutions or insights from the sheaf-theoretic perspective in its right column. Table 6 maps key accomplishments in risk analysis—many of which highlight disciplinary or structural fragmentation—onto corresponding sheaf-theoretic concepts that aim to resolve such fragmentation through formal integration methods, building on the syntheses by Greenberg et al. (2012, 2020).

Across the many domains of risk and decision science explored in Table 5 and the supporting information—from belief elicitation to regulatory design to public risk communication—a recurring pattern is that locally sound and consistent pieces often cannot be integrated, whether the parts are probabilistic beliefs (as in expert elicitation or Bayesian inference), preference orderings (as in public policy deliberation), causal models (as in epidemiology and toxicology), or decentralized actions (as in emergency response or economic regulation) into unified systems that respect desired global criteria of rationality, fairness, or effectiveness. Moreover, as deepfakes increasingly manipulate trusted modalities of communication (Vaccari and Chadwick 2020), structural incoherence becomes not just a technical issue but a deliberate strategy of information attack. What can be done to improve coherence and coordination in decision-making despite these challenges?

A sheaf-theoretic perspective helps both to formalize and diagnose failures in compositionality and to guide the redesign of information structures, incentives, and decision protocols to build coherence despite fragmented and potentially incorrect or untrustworthy local information. The idea of a global section captures the possibility of reconciling local data, views, or actions into a single coherent model or plan. When no global section exists, cohomology helps clarify how and where compositionality fails by localizing structural obstructions to specific cycles or regions of the system. These obstruction patterns often correspond, once interpreted in context, to missing data, incompatible modeling assumptions, strategic distortions, or communication bottlenecks. Such diagnostics can help guide the redesign of decision protocols, risk communication strategies, governance mechanisms, and collective inference processes.

Rather than treating these inconsistencies as inevitable quirks of bounded rationality, the sheaf-theoretic view invites a constructive counterquestion: *What would a coherent global section look like here, and how could we get closer to it in practice?* A powerful approach is to imagine designing a decision aid—such as an AI assistant or a formal decision-optimization model—that acts on one's behalf. What instructions would we give it? Most people would not tell their assistant to use one decision rule when pressured or distracted and a conflicting one when not, or to treat equivalent problems differently based on framing. Instead, we would strive to articulate stable beliefs and preferences that can guide decisions across contexts.

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TABLE 5 | Impossibility results and constructive responses for integrating fragmented information.

Impossibility theorem or limitation	Implications and constructive responses
Risk psychology: Heuristics and Biases and the behavioral-normative reconciliation problem in decision science and behavioral economics (Tversky and Kahneman 1981)	Human judgments about probabilities, preferences, plans, and risks often appear reasonable locally but violate global coherence requirements. This reflects an underlying impossibility: Cognitive heuristics and affective reasoning tend to produce *pseudo-global sections*—judgment patterns that feel internally consistent but violate logical constraints. Rational choice models—such as Bayesian networks (for beliefs) and expected utility frameworks (for preferences)—correspond to valid global sections
Causal modeling: Lucas Critique and the associational-causal gap for causal predictions in economics, statistics, epidemiology, machine learning (Lucas 1976; Pearl 2010; Schwartz and Prins 2024)	Statistical associations in past data often fail to hold following interventions, undermining the reliability of interventional predictions. Invariant Causal Prediction (ICP) (Peter et al. 2016) offers a constructive path forward: It tests whether local causal conditional probability distributions remain stable across diverse environments. If so, they are likely to reflect true causal relationships and can be assembled into a coherent causal network model.
Fault-intolerant distributed control and coordination: FLP Impossibility Theorem (Fischer et al. 1985)	Consensus cannot be guaranteed in asynchronous systems with even one fault. Felber et al. (2025) give a rigorous sheaf-theoretic characterization of when distributed tasks are solvable in asynchronous systems: A task is solvable if and only if a global section exists over a sheaf built from agent knowledge and task constraints. Their method supports both impossibility proofs and practical protocol construction.
Distributed coordination with adversarial communication: Byzantine Generals Problem (Lamport et al. 1982)	Coordination fails with $\geq 1/3$ malicious agents. Trust scaffolding (e.g., cryptographic voting, audit chains) and redundancy support partial recovery. Sheaf theory may help identify where structural repair is needed.
Mechanism design: Myerson–Satterthwaite Theorem (Myerson and Satterthwaite 1983)	No efficient, truthful, budget-balanced mechanism exists for bilateral trade with private values. However, *dynamic mechanisms* (Athey and Segal 2013) can successfully use intertemporal transfers to achieve efficiency, incentive compatibility, and budget balance in repeated settings with private but independent values.
Failures of Collective Action (Game Theory / Public Goods; e.g., Tragedy of the Commons, Prisoner's Dilemma)	Cooperation breaks down in one-shot or weakly monitored settings. Repeated-game Folk Theorems (Fudenberg and Maskin 1986) show cooperation *can* be sustained with memory and incentives. Ostrom (1990, 2000) shows how monitoring, norms, and trust in local institutions overcome collective action failures.
Social choice theory: Arrow's Impossibility Theorem (Arrow 1951)	No aggregation rule satisfies all normative axioms for aggregating preference orderings. However, Harsanyi's (1955) theorem shows aggregation *is* possible (via a utilitarian weighted sum) for utility functions under expected utility maximization.
Gibbard–Satterthwaite Theorem (Voting Theory; Gibbard 1973; Satterthwaite 1975)	Strategy-proofness implies dictatorship if preferences are unrestricted. Domain restrictions, randomized mechanisms, or outcome-contingent commitments can avoid this constraint (Gibbard 1973).
Sen's Liberal Paradox (Welfare Economics; Sen 1970)	Individual rights can block Pareto improvements. Some systems use ethically weighted aggregation or conditional override rules to balance autonomy with collective welfare.
Hylland–Zeckhauser Impossibility Theorem for aggegating beliefs and utilities (Hylland and Zeckhauser 1979)	Aggregating beliefs and utilities separately can yield dictatorship of beliefs. However Pivato (2022) shows that robust, long-run consensus preferences under accumulating evidence can justify utilitarian aggregation without requiring belief aggregation—thus escaping the belief dictatorship constraint.
Nehring's Impossibility Theorem (Collective Judgment; Nehring 2007)	Aggregation can yield group decisions rejected by all individuals. Structured domains (Dietrich and List 2007) and restricted judgment formats enable coherent outcomes.
Mis/Disinformation in opinion dynamics in social networks (Milli 2021; Hansen and Ghrist 2021; Jeong and Shin 2024)	Belief fragmentation, echo chambers, and polarization can persist even among rational agents due to trust-weighted influence, bounded confidence, and early signal distortion. Sheaf Laplacians and influence matrix decompositions identify key agents, pathways, and bottlenecks for targeted interventions (e.g., bridge-building, trust calibration, selective debunking) to mitigate information cascades and improve societal resilience to misinformation.

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TABLE 6 | Risk Analysis accomplishments and corresponding sheaf-theoretic perspectives *Sources*: Greenberg et al. (2012, 2020); this paper.

Risk analysis accomplishment	Sheaf-theoretic perspective
Affect and trust in risk perception	Emotion-driven beliefs (Slovic et al. 2007) form local sections that cannot be glued into a factually coherent global view. Sheaf methods (cohomology and gluing tests) diagnose and localize such failures and suggest structural fixes.
Prospect theory and behavioral biases	Framing effects (Tversky and Kahneman 1981) and time inconsistency (Strotz 1955) yield local preferences that resist global synthesis. Sheaf theory explains these failures of integration (via global section analysis) and reveals structural sources of instability.
Environmental justice (EJ)	Mismatches in geographic scale, metrics, and temporal scope lead to contradictory community-level assessments. Sheaf theory reveals where restriction map misalignments prevent gluing. (Greenberg et al. 2025; Cutter et al. 2004)
Formal uncertainty analysis	Incompatible assumptions across probabilistic models (e.g., expert beliefs or site-specific data) block coherent fusion. Sheaf cohomology detects these structural inconsistencies even when pairwise agreements appear sound.
Multi-objective decision making	Diverse stakeholder preferences can be represented as local sections. Sheaf-theoretic integration (via global section synthesis) finds compromise policies that honor all local priorities. (Keeney and Raiffa 1976)
Regional economic impact modeling	Local CGE and I/O models may conflict across jurisdictional boundaries. Sheaf analysis identifies where structural mismatches (in restriction maps and assumptions) block consistent projections.
Hazard and event modeling	Models with differing exposure assumptions or temporal windows may be locally valid but globally incompatible. Sheaf cohomology localizes these integration obstructions.
Adversarial risk and terrorism modeling	Competing actors may develop internally consistent strategies that cannot be globally reconciled (Bier 2006; Cox 2009). Sheaf theory reveals obstructions due to asymmetric observability or deception (using cohomology on belief structures).
Risk communication	Conflicting frames (e.g., scientific vs. emotional) create misaligned messages across agencies. Sheaf models show how semantic incompatibility in restriction maps blocks coherent public understanding.
Risk-informed regulation	Policy conflicts (e.g., between state and federal models) emerge from incompatible causal assumptions and knowledge (Bier and Lin 2013). Sheaf methods (gluing tests and cohomology) show when structural misalignment prevents regulatory coherence.
Big data and meta-analysis	Pooling studies with heterogeneous exposures or adjustments can fail structurally. Sheaf theory clarifies when integration is feasible (based on gluing conditions and restriction compatibility).
Resilience analytics	Systems may seem robust locally but fail globally under stress. Sheaf theory defines resilience as preservation of a global section under perturbation and uses cohomology to identify latent fragilities (Krasnovsky 2025).
Extreme events & deep uncertainty	Inconsistent or incomplete scenario structures block integration across possible futures (Cox 2012). Sheaf models explain this as a failure of restriction coverage, revealing structural sources of deep uncertainty and potential integration strategies.
Governance and multi-scale decision-making	Misaligned policies across jurisdictions (e.g., local vs. federal) reflect failures in nested restriction maps. Sheaf theory highlights where gluing breaks down across levels.
Synthetic biology and emerging technologies	Modular and incomplete knowledge from lab, ecological, and exposure models often resists integration. Sheaf theory tests whether partial views can be coherently synthesized (via global section conditions).

This thought experiment leads naturally to models grounded in subjective expected utility (SEU) theory and Bayesian decision theory. These frameworks are, in effect, attempts to construct global sections: They define coherent mappings from possible states of the world to beliefs, utilities, and actions. While real human judgments often depart from SEU ideals (as illustrated by framing effects, probability distortions, and time inconsistency), the sheaf-theoretic lens offers a structured way to analyze where and why coherence fails—and how to restore it. In practice, several strategies can help:

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- *Meta-cognitive reflection*. Training analysts or decision-makers to examine whether their judgments across contexts can be reconciled into a consistent whole—analogous to checking whether local sections glue—is a promising educational intervention.
- *Normative modeling*. Using influence diagrams, Bayesian networks, or optimization models to formalize what a global section would be in a given decision setting can clarify tensions and trade-offs that were previously implicit or conflicting.
- *AI-assisted coherence checking*. Decision support systems or personal decision aids could use sheaf-theoretic diagnostics to flag inconsistencies across a user's preferences, risk assessments, or plans—similar to how spellcheckers flag inconsistencies in spelling.
- *Institutional design*. At a broader level, policies and procedures can be evaluated based on whether they enable the parts of a system (e.g., agencies, models, stakeholders) to work together coherently—or whether they institutionalize misalignment and incoherence.

Additional strategies include designing systems to support multi-scale modeling and abstraction (to locate coherence at higher levels of description); using constraint-based formulations and optimization to guide stakeholder negotiation and compromise; implementing coherence diagnostics within analytic and organizational workflows; and enabling deliberative fusion of local models using structured, sheaf-compatible methods. These approaches can help build systems that not only detect but also resolve incoherence arising from the attempted fusion of fragmented inputs.

Ultimately, the sheaf-theoretic perspective does more than clarify sources of failure in coordination or coherence. It suggests a roadmap for improvement: Think globally, model locally, and test for coherence. If a system lacks a global section, that is not the end of the analysis. It is the beginning of a principled search for better-aligned information, preferences, and actions across scales, stakeholders, and time. As risk challenges grow more distributed and interdependent, the search for valid global sections may offer a valuable perspective for structuring analytic tools, policy conversations, and risk management decision-support systems.

5 | Discussion: Sheaf Theory and Progress in Risk Analysis

Over the past four decades, risk analysis has matured from an applied toolkit for decision support into a multidisciplinary field with a broad conceptual and methodological foundation. Hallmarks of this evolution include rigorous quantification of uncertainty, widespread use of probabilistic risk assessment (PRA), incorporation of expert judgment, integration of behavioral insights, attention to risk communication and perception, and institutionalization of structured decision-making frameworks. Foundational papers (e.g., Fischhoff et al. 1978; Kaplan and Garrick 1981) emphasized the value of probabilistic reasoning under uncertainty and formal structures for risk-informed decision-making. Later advances incorporated Bayesian methods, influence diagrams, value-of-information techniques, and more nuanced approaches to risk perception, stakeholder engagement, and policy relevance.

Recent work by Terje Aven and colleagues has also emphasized the central role of epistemic structure in risk analysis. The approach highlights that meaningful risk characterization depends not only on judgments of the likelihood and severity of outcomes, but also on the strength, reliability, and limits of the underlying knowledge. The framework draws a critical distinction between *risk* (C,U), where C stands for consequences or outcomes and U for uncertainty about them, as a concept (reflecting real-world activities and uncertainty about consequences) and the *characterization* of risk produced by a risk assessment, denoted (C',Q,K), based on a knowledge base K; a description of the relevant uncertainties, Q; and a description of consequences, C'. This distinction is critical as any characterization is based on some knowledge which could be wrong (e.g., based on incomplete, incorrect, or inconsistent data, information, assumptions, analysis, models, statistical tests, and so forth). Important aspects of risk could be overlooked in the assessment—for example, events not identified in C'. Q is a *description* of the uncertainties, but that may be subject to imprecision and bias issues. This framework implies a need for epistemic humility, acknowledging that the risk characterization (C', Q, K) may not capture all aspects of the risk (C, U) being considered. From a sheaf-theoretic perspective, such formulations might be interpreted as local sections over different domains or scenarios, with the knowledge component K playing the role of a restriction map determining how local assessments of risk relate to one another and whether they can be coherently integrated. Aven's notation points toward a structural, system-aware understanding of how local beliefs and uncertainties must align to support valid global inferences and risk science. Sheaf-theoretic tools offer a way to make this structure explicit, testable, and potentially computable.

Yet despite this impressive progress, certain structural challenges persist across domains: How to integrate fragmented data and models, how to resolve conflicts among expert and stakeholder beliefs, how to identify when group decisions are incoherent or vulnerable to failure, and how to design systems that maintain coordination and rationality in the face of partial knowledge, delays, or strategic incentives. We propose that many of these challenges share a common underlying structure: They are *failures of compositionality*. We have argued that sheaf theory provides a useful mathematical language and conceptual framework for articulating when and how such failures arise, and for organizing corrective strategies. Much as Bayesian networks, influence diagrams, and decision trees provided unifying tools in earlier generations of decision and risk analysis, sheaf-theoretic methods might offer new tools for identifying and overcoming integration problems in distributed decision systems.

From this perspective, key accomplishments in risk analysis can be reinterpreted as early responses to specific forms of compositionality failure: The use of expert elicitation and structured judgment protocols as partial attempts to glue fragmented knowledge; the emphasis on stakeholder engagement and risk communication as attempts to align diverse perspectives; the development of scenario analysis and influence diagrams as tools for exploring conditional structure and coherence.

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The sheaf-theoretic perspective does not replace these methods, but provides a coherent meta-framework for understanding their structure, scope, and limitations. It also invites a new generation of tools: Cohomological diagnostics for inconsistency, global section search for optimal coordination, modular mechanism design for incentive alignment, and influence structure optimization for managing belief dynamics. In this way, sheaf theory complements and advances the core aims of risk analysis—improving decisions under uncertainty by integrating diverse sources of knowledge, value, and action.

The value of this structural perspective becomes especially clear when placed alongside the historical trajectory of the field, as in Table 6. This aligns with calls to revisit and strengthen foundational assumptions and unify diverse perspectives in risk science (Aven and Zio 2014). The pairings in Table 6 suggest how each major advance in risk analysis in recent decades (Greenberg et al. 2012, 2020)—from multi-objective tradeoff analysis to resilience modeling—might benefit from sheaf theory concepts and tools that help organize, align, and interrogate complex systems of information and action.

As risk analysis moves into its fifth decade, the central challenges it faces—in climate policy, AI safety, global health, energy transition, and synthetic biology—are not only technical, but epistemic and organizational. They concern how to coordinate understanding, decision-making, and responsibility across fractured and rapidly evolving systems. As noted by Aven and Zio (2014), foundational challenges in risk assessment often arise from discipline-specific silos and mismatched assumptions across actors and levels. We have proposed that sheaf theory concepts and methods can help to identify, localize, and potentially resolve these incompatibilities, making sheaf theory a useful candidate foundation for future unification efforts.

6 | Conclusions

From the interpretation of probabilistic beliefs and preferences to the coordination of team decisions, regulatory processes, and public policy, modern risk analysis is replete with problems of integration: how to reconcile diverse local models, judgments, or behaviors into a coherent and decision-relevant whole. This paper has argued that many of these integration problems—although perhaps often viewed as domain-specific anomalies or communication failures—are, in fact, instances of a deeper mathematical phenomenon: The failure of compositionality. When locally consistent parts cannot be assembled into a globally consistent structure, the result may be cognitive biases, unreliable expert elicitation, flawed causal inference, inconsistent regulation, or failed collective action—not because of irrationality, but because the underlying information structures do not support a coherent synthesis.

Sheaf theory provides a rigorous, general framework for analyzing and, where possible, repairing such failures. By representing local data, assumptions, and inferences as mathematical objects associated with specific domains or contexts, and defining how they must agree on overlaps, sheaf models can formally test whether a consistent global picture exists and identify the reasons when one does not. As shown in the examples throughout Section 4, this approach applies across levels of analysis—from individual beliefs and preferences to institutional decision protocols and emergent societal dynamics.

We believe that the potential value of sheaf theory for advancing risk analysis comes not only from its conceptual generality, but also from its implications for practice. It offers decision analysts, risk modelers, and policymakers ways to understand and correct misalignments in institutional and organizational designs and incentives, information structures, communication flows, and integration of local perceptions and plans. It provides a set of tools for anticipating and mitigating failures in integration, coordination, and cooperation before they lead to costly errors or public mistrust. It suggests new ways to build modular, diagnosable, and flexible models. And it points toward the possibility of future AI systems that can reason about inconsistency, ambiguity, and decentralized information with more transparency and insight than current systems allow.

Sheaf theory also complements probabilistic and causal modeling by offering structural insight into the feasibility of consistent integration across distributed data sources. In the presence of mis- or disinformation, this lens becomes even more critical. Rather than treating untrustworthy information as random noise, sheaf-based methods allow risk analysts to consider whether adversarial or strategic distortions, semantic mismatches, or conflicting encodings make coherent global understanding impossible under the current system structure—and to identify where targeted redesigns (e.g., new trust bridges, standardized interpretations, or semantic harmonization) might restore coherence. Just as probability theory and optimization once transformed the tools and language of risk analysis, so too may sheaf theory help to provide a next-generation framework—particularly in contexts where integration across models, perspectives, or scales is essential but hard to achieve (Aven and Zio 2014). It may not only help us to reason better under uncertainty and with fragmented information from multiple, not necessarily trustworthy, sources; but also equip us to diagnose and improve the structure of information collection, communication, decision-making, and reasoning itself in uncertain, complex, and distributed risk systems.

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I used OpenAI's ChatGPT throughout the preparation of this paper in generating the tables and appendices and in helping to draft initial versions of successive outlines and text. The author is responsible for all aspects of the final text. I believe that such human-AI collaborations

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highlight the potential of AI to help increase productivity and advance the quality of research.

Conflicts of Interest

The author declares no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting: risa70206-sup-0001-OnlineSupplement-1.docx **Supporting**: risa70206-sup-0001-OnlineSupplement-2.docx

Appendix

R code for Sheaf Theory Model of Opinion Dynamics in Figure 2

```
# Load required package
library(Matrix)

  # Step 1: Define network nodes and edges

nodes ← c("A", "B", "C", "D")
edges ← list(
c("A", "B"),
c("B", "C"),
c("C", "D"),
c("D", "A"),
c("A", "C")
)

  # Step 2: Define symmetric trust weights (restriction maps)

restriction ← list(
"A_B" = 0.9, "B_A" = 0.9,
"B_C" = 0.7, "C_B" = 0.7,
"C_D" = 0.8, "D_C" = 0.8,
"D_A" = 0.6, "A_D" = 0.6,
"A_C" = 0.5, "C_A" = 0.5
)

  # Step 3: Helper function for consistent edge keys

edge_key ← function(u, v) {
paste(sort(c(u, v)), collapse = "_")
}

  # Step 4: Build D0 matrix

n_edges ← length(edges)
n_nodes ← length(nodes)
D0 ← matrix(0, nrow = n_edges, ncol = n_nodes,
dimnames = list(paste0("e", 1:n_edges), nodes))
for (i in seq_along(edges)) {
u ← edges[[i]][1]
v ← edges[[i]][2]
key ← edge_key(u, v)
trust ← restriction[[key]]
if (is.null(trust)) stop(paste("Missing trust value for edge", u, v))
D0[i, u] ← trust
D0[i, v] ← -trust
}

  # Step 5: Compute sheaf Laplacian L = t(D0) %*% D0

L ← t(D0) %*% D0

  # Step 6: Define initial beliefs with misinformation at node A

beliefs ← matrix(c(1.5, 0.8, 0.0, 0.2), nrow = 4, ncol = 1,
dimnames = list(nodes, "belief"))

  # Step 7: Run belief dynamics

eta ← 0.1
T ← 20
X_mis ← matrix(NA, nrow = T + 1, ncol = length(nodes),
dimnames = list(0:T, nodes))
X_mis[1,] ← beliefs[, 1]
for (t in 1:T) {
X_mis[t + 1,] ← X_mis[t,]—eta * (L %*% X_mis[t,])
```


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```
}

  # Step 8: Plot belief trajectories with custom colors

time_steps ← 0:T
curve_colors ← c("red", "orange", "green", "blue")
matplot(time_steps, X_mis, type = "l", lty = 1, lwd = 2,
col = curve_colors, xlab = "Time Step", ylab = "Belief Value",
main = "Belief Dynamics with Misinformation at Node A")
legend("topright", legend = colnames(X_mis), col = curve_colors,
lty = 1, lwd = 2)
```


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